H2O: The Versatile Tool for Deep Learning

The H2O project developed by H2O.ai provides users tools for data analysis, allowing them to fit thousands of potential models when trying to discover patterns in data. It is a very versatile tool since it is supported by various programming languages like R, Python and MATLAB.

Deep learning is a superset of the artificial neural network architecture and is gradually becoming an essential tool for data analysis and prediction. Leading programming languages like R, Python and MATLAB provide powerful tools and support for the implementation of data analysis using deep learning. Among such different tools, both TensorFlow and H2O have supportive packages in R and Python. Hence, both these languages are steadily becoming indispensable for deep learning and data analysis.

Deep learning with H2O

The main objective of H2O.ai is to add intelligence to business. This works on the principle of deep learning and computational artificial intelligence. H2O provides easy solutions for data analysis in the financial services, insurance and healthcare domains, and is gradually proving itself as an efficient tool for solving complex problems.

This deep learning tool follows the multi-layer feed forward neural networks of predictive models, and uses supervised learning models for regression analysis and classification tasks. To achieve process-parallelism over large volumes of data distributed over a cluster or grid of computational nodes, it uses the MapReduce algorithm. With the help of various mathematical algorithms, MapReduce divides a task into small parts and assigns them to multiple systems. H2O is scalable from small PCs to multi-core servers and even to multi-core clusters. To prevent over-fitting, different regularisation techniques are used. Commonly used approaches are L1 and L2 (Lasso and Ridge). Other than these approaches, H2O uses the dropout, HOGWILD! and model averaging methods also. For non-linear activation functions, H2O uses the Hyperbolic tangent, Rectifier Linear and Maxout functions. The performance of each function depends on the operational scenarios and there is no one best rule to base one's selection upon. For error estimation, this model uses either one of the Mean Square Error, Absolute, Huber or Cross Entropy functions. Each of these loss functions are strongly associated with a particular data distribution function and are used accordingly.